A Generative Model for Atomic Static Dipole Polarizability

Descriptor Design, Fitting Methodology, and Validation (Z = 1–120)

Formation Engine — Atomistic Simulation Library

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Abstract

We present a 13-parameter generative model for the scalar isotropic static dipole polarizability $\alpha(Z)$ of neutral atoms, covering $Z = 1$–118 with an experimental extension to $Z = 120$. The model requires no external data files at runtime: all inputs are derived from the atomic number Z alone via closed-form descriptor functions (covalent radius, period, block, electronegativity proxy, softness). Parameters are fitted offline against 118 reference values from Miller (1990) and Schwerdtfeger & Nagle (2019) using iteratively-reweighted Huber regression with group-monotonicity regularisation. The overall RMS error is 19.1% across all 118 elements and <16% for the commonly-simulated $Z = 1$–54 subset. The model feeds directly into the Thole-damped SCF induced-dipole solver that computes polarization energies and forces in molecular dynamics.

1 Model Equation

The polarizability of element Z is predicted by

$$\boxed{\alpha(Z) \;=\; \underbrace{c_b(Z)\, k_p(Z)\, r_{\mathrm{cov}}(Z)^3\, \frac{1 + a_{\mathrm{soft}}\, s(Z)}{\chi(Z)^{b_\chi}}}_{\alpha_{\mathrm{base}}} \cdot \underbrace{g_{\mathrm{shell}}(Z)}_{\text{closure gate}} \cdot \underbrace{g_{\mathrm{rel}}(Z)}_{\text{relativ. stiffening}}} \tag{1}$$

where the terms are:

Symbol	Meaning	Source
$r_{\mathrm{cov}}(Z)$	Covalent radius (Å)	Pyykkö & Atsumi (2009)
$s(Z)$	Softness descriptor (dimensionless)	Eq. (**??**)
$\chi(Z)$	Electronegativity proxy	Eq. (**??**)
$k_p(Z)$	Period scaling factor ($p = 1 \ldots 7$)	Fitted
$c_b(Z)$	Block correction ($b \in \{s, p, d, f\}$)	Fitted
a_{soft}	Softness coupling coefficient	Fitted
b_χ	Electronegativity suppression exponent	Fitted
$g_{\mathrm{shell}}(Z)$	Shell-closure suppression (Eq. **??**)	Fitted
$g_{\mathrm{rel}}(Z)$	Relativistic stiffening (Eq. **??**)	Fitted

The model has **15 free parameters**: 7 period coefficients $k_p^{(1)} \ldots k_p^{(7)}$, 4 block corrections $c_b^{(s)}, c_b^{(p)}, c_b^{(d)}, c_b^{(f)}$, 2 global exponents $a_{\mathrm{soft}}, b_\chi$, and 2 gate amplitudes $c_{\mathrm{shell}}, a_{\mathrm{rel}}$.

1.1 Physical Motivation

The leading-order physics is the *volume scaling law*: polarizability grows as the cube of the electron-cloud radius. For neutral atoms the covalent radius r_{cov} is the best single-descriptor proxy for the mean valence-shell extent. The remaining terms correct for effects that r_{cov}^3 alone cannot capture:

1. **Period scaling (k_p):** Outer-shell electrons in higher periods are more diffuse per unit radius due to additional radial nodes. The fitted k_p values decrease monotonically from period 1 to period 7, reflecting the diminishing "volume efficiency" as shielding increases.

2. **Electronegativity suppression (χ^{b_χ}):** Electronegative atoms hold their electron cloud tighter, reducing deformability below what r_{cov}^3 predicts. The denominator χ^{b_χ} implements this as a power-law suppression.

3. **Softness coupling ($1 + a_{\text{soft}}\, s$):** The softness descriptor s measures how loosely-held the valence electrons are (high for alkali metals, low for noble gases). The fitted $a_{\text{soft}} < 0$ *reduces* α for soft atoms—a counter-intuitive result that compensates for the period scaling already being fit to alkali metals.

4. **Block correction (c_b):** Transition metals (d-block) and lanthanides/actinides (f-block) have polarisable sub-valence electrons ($3d$, $4f$) whose spatial extent is not captured by r_{cov}. The block multiplier absorbs this orbital-type bias.

2 Shell-Closure Gate

2.1 Why Noble Gases Blow Up (and How to Fix Without Hacks)

Noble gases are closed-shell atoms: their electron clouds are stiff and compact. Polarizability is comparatively low and follows a *different curvature* than smooth volume scaling—shell closure breaks the monotonic "bigger radius ⇒ more polarisable" intuition that the base model assumes.

Before v2.6.2, the noble-gas RMS was 36.7%, with Ne at +85%. The existing χ-based denominator in α_{base} partially corrected for this (by assigning noble gases a high effective electronegativity), but was insufficient because r_{cov}^3 still dominates.

2.2 The Shell-Closure Gate

Noble gases should not be "smoothed" into the same curve as their neighbours. They must be *gated*: a multiplicative penalty that activates at or near shell closure.

We define a *shell-closure proximity* descriptor:

$$\phi(Z) = \exp\left(-\frac{(18 - g(Z))^2}{2\sigma_{\text{shell}}^2}\right) \tag{2}$$

where $g(Z)$ is the IUPAC group number ($g = 18$ for noble gases) and $\sigma_{\text{shell}} = 1.8$. This Gaussian is centred on group 18:

- Noble gases ($g = 18$): $\phi = 1.0$ (full closure)
- Halogens ($g = 17$): $\phi \approx 0.86$ (near-closure)
- Chalcogens ($g = 16$): $\phi \approx 0.55$ (moderate)

- $g \leq 14$: $\phi < 0.15$ (negligible)
- f-block ($g = 0$): $\phi = 0$ by convention

The gate is then:

$$g_{\text{shell}}(Z) = 1 - c_{\text{shell}} \cdot \phi(Z) \tag{3}$$

where c_{shell} is a single fitted amplitude. The fitted value $c_{\text{shell}} = 0.328$ means noble-gas polarizabilities are suppressed by $\sim 33\%$ relative to the base model, while halogens get a $\sim 28\%$ suppression—physically correct, as the halogen anion X^- (iso-electronic with the adjacent noble gas) has far higher polarizability than neutral X.

Design principle: gate, don't smooth. Smoothing residuals across a window that includes noble gases will try to "average out" closure physics, creating a correction curve that looks pretty and is physically wrong. Noble gases are a *regime boundary*: the model must acknowledge the discontinuity in electronic structure at shell closure, not paper over it.

3 Relativistic Stiffening

3.1 Why Mercury Spikes (and What Fixes It)

Mercury ($Z = 80$) is the poster child for relativistic orbital contraction: its $6s$ orbital contracts and stabilises, reducing participation in polarizability and making Hg unusually "inert" compared to naive trend expectations. This is the same physics behind liquid mercury and its anomalous bonding.

Before v2.6.2, the model treated Hg like a normal heavy metal with a fat, squishy outer cloud: predicted 7.79 Å^3 vs. reference 5.70 Å^3 (+37%).

3.2 The Relativistic Stiffening Factor

Relativistic contraction is *not* monotonic with Z—it peaks at specific electronic configurations. We identify two hotspots:

1. $6s$ **inert-pair region** ($Z \approx 80$): Au(79), Hg(80), Tl(81) all have anomalously compact $6s$ orbitals due to strong nuclear shielding.
2. $7p_{1/2}$ **spin–orbit region** ($Z \approx 115$): The superheavy p-block (Nh–Og) experiences extreme j-dependent orbital splitting. The $7p_{1/2}$ subshell contracts strongly, reducing the effective polarisable volume.

We model these as a sum of two Gaussians:

$$f_{\text{rel}}(Z) = \exp\left(-\frac{(Z-80)^2}{200}\right) + \frac{1}{2}\exp\left(-\frac{(Z-115)^2}{50}\right) \tag{4}$$

and the gate is:

$$g_{\text{rel}}(Z) = 1 - a_{\text{rel}} \cdot f_{\text{rel}}(Z) \tag{5}$$

The first Gaussian ($\sigma = 10$) yields $f_{\text{rel}} \approx 1.0$ at Hg, ≈ 0.7 at Au/Tl, and decays to < 0.1 by Gd(64) and Po(84). The second Gaussian ($\sigma \approx 5$) peaks at 0.5 for Mc(115) and covers Nh(113)–Og(118).

The fitted amplitude $a_{\mathrm{rel}} = 0.054$ is physically small, consistent with the expectation that relativistic effects are a *perturbative correction* to the volume-scaling base model, not a dominant term. The correction reduces Hg's error from +37% to +34% and improves the overall period-6 RMS from 15.4% to 14.1%.

Why not a sigmoid? An earlier version used a sigmoid $\sigma(Z) = 1/(1 + e^{-(Z-76)/8})$. This failed ($a_{\mathrm{rel}} = 0$ in all fits) because it applies uniform dampening to *all* heavy atoms: Os, Ir, and Pt—which fit well without correction—were degraded more than Hg was improved. The double-Gaussian targets only the physical hotspots, allowing the fitter to find a non-zero optimum.

Hg as a calibration anchor. If Hg's error improves, Au, Rn, and the $7p$ "inert pair" elements (Nh–Og) typically improve as well, because the same $6s/7p_{1/2}$ contraction physics underlies all of them. Hg should therefore be monitored as a leading indicator in future model revisions.

4 Descriptor Definitions

All descriptors are pure functions of Z. No external database is loaded at runtime.

4.1 Covalent Radius $r_{\mathrm{cov}}(Z)$

Tabulated for $Z = 1$–118 from Pyykkö & Atsumi, *Chem. Eur. J.* **15**, 186 (2009). Stored as a compile-time constant array. Units: Å.

4.2 Period and Block

Standard IUPAC periodic table assignment:

$$p(Z) \in \{1, 2, 3, 4, 5, 6, 7\}, \qquad b(Z) \in \{s,\, p,\, d,\, f\}$$

Lanthanides ($Z = 57$–71) and actinides ($Z = 89$–103) are assigned $b = f$. Group 0 (no IUPAC group number) is used for the f-block insets; all other elements have standard group assignments $g = 1$–18.

4.3 Active Valence Electron Count $n_{\mathrm{val}}(Z)$

$$n_{\mathrm{val}}(Z) = \begin{cases} g & g \in \{1, 2\} \quad (s\text{-block}) \\ g - 10 & g \in \{13, \ldots, 18\} \quad (p\text{-block}) \\ g & g \in \{3, \ldots, 12\} \quad (d\text{-block}) \\ 3 & Z \in [57, 71] \cup [89, 103] \quad (f\text{-block}) \end{cases} \tag{6}$$

f-block design decision. Lanthanide and actinide $4f/5f$ electrons are spatially contracted and behave as core-like. Setting $n_{\mathrm{val}} = 3$ for all 30 elements prevents an artificial monotonic gradient in $s(Z)$ across each series. The actual variation in polarizability within the lanthanide contraction is absorbed by $c_b^{(f)}$ and $r_{\mathrm{cov}}(Z)^3$. This invariant is enforced by test T1.

4.4 Maximum Period Valence $n_{\max}(p)$

$$n_{\max}(p) = \begin{cases} 2 & p = 1 \\ 8 & p \in \{2, 3\} \\ 18 & p \in \{4, 5\} \\ 32 & p \in \{6, 7\} \end{cases} \tag{7}$$

4.5 Electronegativity Proxy $\chi(Z)$

An Allred–Rochow-style estimate:

$$\chi(Z) = \frac{n^*_{\text{val}}(Z)}{r_{\text{cov}}(Z)^2} \times 0.359 \tag{8}$$

where the effective valence count n^*_{val} equals $n_{\text{val}}(Z)$ for all elements except noble gases ($g = 18$), which use $n^*_{\text{val}} = n_{\max}(p)$ (full shell). The normalisation constant 0.359 is chosen so that carbon ($n_{\text{val}} = 4$, $r_{\text{cov}} = 0.75$ Å) gives $\chi_{\text{C}} \approx 2.55$ on the Pauling scale.

Noble gas electronegativity. Setting $n^*_{\text{val}} = n_{\max}$ for group 18 produces artificially high χ, which drives α down through the denominator of Eq. (**??**). This is physically correct: noble gases have compact, tightly-bound closed shells and small polarizabilities despite moderate covalent radii.

4.6 Softness $s(Z)$

$$s(Z) = \frac{n_{\text{val}}(Z) \;/\; n_{\max}\big(p(Z)\big)}{\chi(Z)} \tag{9}$$

Softness is high for diffuse, easily-deformable electron clouds (alkali metals: $s_{\text{K}} \approx 5.5$) and low for tight shells (fluorine: $s_{\text{F}} \approx 0.3$; neon: $s_{\text{Ne}} \approx 0.03$).

5 Fitting Methodology

5.1 Reference Data

118 reference polarizabilities from two sources:

- **Miller (1990):** gas-phase experimental values for $Z = 1$–54 (JACS 112, 8533).
- **Schwerdtfeger & Nagle (2019):** relativistic DHF/CCSD(T) values for $Z = 21$–118 (Mol. Phys. 117, 1200).

Each entry carries a fitting weight $w_i \in [0.1, 2.0]$ encoding reliability: $w = 2.0$ for high-priority simulation elements (H, C, N, O, S), $w = 0.3$ for radioactive elements with theory-only values.

5.2 Loss Function

The objective is a weighted Huber loss in log-space plus a smoothness penalty:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{N} w_i \, \mathcal{H}_\delta\Big(\ln \hat{\alpha}_i - \ln \alpha_i^{\text{ref}}\Big) \;+\; \eta \sum_{(a,b)\in\mathcal{P}} \mathcal{H}_\delta\Big(\Delta_{ab}^{\text{pred}} - \Delta_{ab}^{\text{ref}}\Big) \tag{10}$$

where $\mathcal{H}_\delta(x) = \frac{1}{2}x^2$ for $|x| \le \delta$ and $\delta(|x| - \frac{1}{2}\delta)$ otherwise, with $\delta = 0.3$. The set $\mathcal{P}$ contains pairs of elements that are adjacent in the same periodic-table group, sorted by period; $\Delta_{ab} = \ln \alpha_b - \ln \alpha_a$ enforces monotonicity of trends down each group.

Why log-space. Polarizabilities span three orders of magnitude ($\alpha_{\mathrm{He}} = 0.205$ Å^3 to $\alpha_{\mathrm{Cs}} = 59.4$ Å^3). Log-space Huber loss treats a 50% error on helium as seriously as a 50% error on caesium.

5.3 Optimisation: Iterative Reweighted Coordinate Descent

The 13 parameters are fitted by a two-loop scheme:

1. **Outer loop** (up to 30 rounds): after each inner convergence, update element weights w_i based on residuals:

$$w_i \leftarrow \min\bigl(w_{\max},\ w_i\bigl(1 + \alpha_{\mathrm{rw}} \cdot \max(0,\ |\epsilon_i|/\tau - 1)\bigr)\bigr)$$

   where $\epsilon_i = \ln \hat{\alpha}_i - \ln \alpha_i^{\mathrm{ref}}$, $\tau = 0.20$ is the tolerance target, $\alpha_{\mathrm{rw}} = 0.3$, and $w_{\max} = 10$.

2. **Inner loop** (up to 80 sweeps): coordinate descent over all 13 parameters. Each parameter is optimised by a multi-pass grid search (4 passes $\times$ 100 grid points, geometrically refining the search window).

Best-checkpoint selection. The weighted loss $\mathcal{L}$ monotonically increases across outer iterations (because weights grow), but the *unweighted* RMS error may degrade after the optimal reweighting round. The fitter tracks the unweighted RMS at every outer iteration and returns the parameters from the checkpoint with the lowest unweighted RMS, not the final iterate.

6 Fitted Parameters (v2.6)

Parameter	Value	Physical role
$k_p^{(1)}$	20.820	Period 1 (H, He)
$k_p^{(2)}$	5.381	Period 2 (Li–Ne)
$k_p^{(3)}$	5.117	Period 3 (Na–Ar)
$k_p^{(4)}$	3.956	Period 4 (K–Kr)
$k_p^{(5)}$	3.728	Period 5 (Rb–Xe)
$k_p^{(6)}$	2.919	Period 6 (Cs–Rn)
$k_p^{(7)}$	2.665	Period 7 (Fr–Og)
a_{soft}	−0.680	Softness coupling
b_χ	0.225	Electronegativity exponent
$c_b^{(s)}$	1.464	s-block multiplier
$c_b^{(p)}$	0.973	p-block multiplier
$c_b^{(d)}$	1.605	d-block multiplier
$c_b^{(f)}$	1.745	f-block multiplier
c_{shell}	0.328	Shell-closure suppression amplitude
a_{rel}	0.054	Relativistic stiffening amplitude

The monotonic decrease $k_p^{(1)} > k_p^{(2)} > \cdots > k_p^{(7)}$ reflects increasing nuclear shielding: each additional shell of core electrons makes the volume-to-polarizability conversion less efficient. The d-block multiplier $c_b^{(d)} \approx 1.6$ confirms that partially-filled d shells contribute significant polarisable density beyond what r_{cov}^3 alone predicts. The shell-closure amplitude $c_{\text{shell}} \approx 0.33$ suppresses noble-gas predictions by a third, while the relativistic amplitude $a_{\text{rel}} \approx 0.05$ provides a targeted $\sim 5\%$ correction at the Hg hotspot.

7 Validation Results

7.1 Global Accuracy

Metric	**Z = 1–118**	**Z = 1–54**
RMS error	17.5%	14.8%
Max error	64.5% (Lv)	57.4% (Ne)
Monotonicity breaks	10/70 pairs	5/35 pairs
Elements >20% error	23/118	8/54

7.2 Per-Block RMS

Block	**RMS**	**Physical note**
s-block	19.0%	Alkali/alkaline-earth: large α, well constrained
p-block	19.9%	Noble-gas suppression now active (g_{shell})
d-block	15.5%	Transition metals: $c_b^{(d)}$ absorbs d-shell effect
f-block	16.0%	Lanthanides/actinides: fixed $n_{\text{val}} = 3$ works well

7.3 Improvement from v2.6.0 to v2.6.2

Regime	**v2.6.0**	**v2.6.2**	**Improvement**
Overall RMS	19.1%	17.5%	−8.4%
Noble gases RMS	36.7%	27.3%	−25.6%
Halogens RMS	24.1%	14.2%	−41.1%
Period 6 RMS	15.4%	14.1%	−8.4%
Hg error	+36.6%	+34.0%	−7.1%
Ne error	+85.0%	+57.4%	−32.5%
Max error	85.0%	64.5%	−24.1%

7.4 Known Failure Modes

1. **Neon** ($\alpha_{\text{ref}} = 0.396$ Å^3, error +85%): Ne has an anomalously small polarizability relative to its covalent radius. The model's χ-based suppression is insufficient because $r_{\text{cov}}(\text{Ne}) = 0.67$ Å overestimates the true electron-cloud extent for this closed-shell atom.

2. **Nobelium** ($\alpha_{\text{ref}} = 17.5$ Å^3, error +48%): No has strong relativistic contraction of the $7s$ orbital that reduces α below the r_{cov}^3 expectation. Our non-relativistic descriptors cannot capture this.

3. **Superheavy p-block** (Lv, Ts: errors >40%): Extreme spin–orbit coupling in the $7p_{1/2}$ / $7p_{3/2}$ manifold changes the effective orbital extent. Relativistic CCSD(T) reference values carry significant uncertainty themselves.

8 Integration with the SCF Polarization Solver

The alpha model provides the atomic polarizabilities $\alpha_i = \alpha(Z_i)$ that enter the Applequist–Thole SCF equations:

$$\boldsymbol{\mu}_i = \alpha_i \, \mathbf{E}_i^{\mathrm{loc}} \tag{11}$$

$$\mathbf{E}_i^{\mathrm{loc}} = \mathbf{E}_i^{\mathrm{perm}} + \sum_{\substack{j \neq i \\ r_{ij} > r_{\mathrm{excl}}}} f(u_{ij}) \, \mathbf{T}_{ij} \, \boldsymbol{\mu}_j \tag{12}$$

where $\mathbf{T}_{ij}$ is the dipole–dipole interaction tensor and $f(u)$ is the Thole exponential damping function:

$$f(u) = 1 - \left(1 + \tfrac{a\,u}{2}\right) e^{-au}, \qquad u_{ij} = \frac{r_{ij}}{(\alpha_i \, \alpha_j)^{1/6}}, \qquad a = 2.6 \tag{13}$$

Unit convention. All electric fields ($\mathbf{E}^{\mathrm{perm}}$ from charges, $\mathbf{T}_{ij}\boldsymbol{\mu}_j$ from dipoles) are computed in Gaussian units ($e/\text{Å}^2$) without the Coulomb constant k_e. The constant $k_e = 332.0636$ kcal $\cdot$ Å/(mol $\cdot$ e^2) appears only in the energy formula:

$$U_{\mathrm{pol}} = -\frac{1}{2} \, k_e \sum_{i=1}^{N} \boldsymbol{\mu}_i \cdot \mathbf{E}_i^{\mathrm{perm}} \tag{14}$$

Exclusion radius. Pairs with $r_{ij} < r_{\mathrm{excl}} = 1.6$ Å are excluded from both the permanent-charge driving field and the dipole–dipole tensor. This is equivalent to standard 1-2/1-3 exclusions in AMOEBA without requiring bond topology.

9 Experimental Extension: Z = 119–120

Elements 119 (ununennium, Uue) and 120 (unbinilium, Ubn) have not been synthesised in measurable quantities. An optional extension module (`alpha_model_ext.hpp`) provides:

- **Literature values**: $\alpha_{119} = 165$ Å^3, $\alpha_{120} = 109$ Å^3 from 4-component DHF calculations (Schwerdtfeger & Nagle, 2019).
- **Generative prediction**: the standard model equation using extrapolated period-8 descriptors ($r_{\mathrm{cov}}(119) = 2.60$ Å, $r_{\mathrm{cov}}(120) = 2.21$ Å from Pyykkö 2015). The period-7 coefficient $k_p^{(7)}$ is reused as a floor.

The generative predictions ($\alpha_{119}^{\mathrm{model}} \approx 103$ Å^3, $\alpha_{120}^{\mathrm{model}} \approx 52$ Å^3) underestimate the literature values, confirming that relativistic $8s$-contraction effects would require a period-8-specific $k_p^{(8)}$ coefficient that cannot be fitted without training data.

10 Software Architecture

File	Role
`atomic_descriptors.hpp`	Pure-function descriptor layer (r_{cov}, χ, s, period, block)
`alpha_model.hpp`	Model equation + baked parameters (13 coefficients)
`alpha_model_ext.hpp`	Optional Z=119–120 extension
`polarization_scf.cpp`	Thole-damped SCF solver (consumes α_i)
`fit_alpha_model.cpp`	Offline fitter (reads CSV, writes JSON + updates baked params)
`polarizability_ref.csv`	Reference data (118 entries, not shipped to production)
`test_alpha_model.cpp`	Descriptor invariants (T1–T4, 56 checks)
`test_nuclear_chemical.cpp`	Phase 6 validation (F1–F5, 24 checks)

Key design constraint: the descriptor layer and model equation are *read-only at runtime*. All tuning happens offline in the fitter; the production binary contains only the baked 13-parameter struct.

References

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